Mitsue Forest → Energy: A Workforce-Led 25-Year Plan

> **"The forest our ancestors planted — the power that sustains the village they built"**
> 先人が植えた森が、今、村を支える力になる。

1. The core idea

The bottleneck in Mitsue's forest is **not the trees — it is the crew and the machines**. 御杖村森林組合 (Mitsue Kanko) today manages only a fraction of the village's **7,051 ha** of forest, most of it as subsidised thinning rather than full harvest.

This project's purpose is to **double — and over time triple — the cooperative's workforce and mechanise it** (harvester + forwarder, chipper, drying). Energy revenue from the harvested sugi pays for that crew and equipment; the larger crew harvests more; more harvest feeds a larger biomass plant. It is a **self-funding loop**, the same model proven in Shimokawa and Nishiawakura.

2. Where the cooperative is today

Item	Current	Source
Village forest area	~7,051 ha (>90% private, sugi/hinoki postwar plantations)	Co-op report
Annual thinning	~66 ha/yr	Co-op report
Raw material recovered	only ~120–300 m³/yr	Co-op report
Crew	small, young (avg age ~33)	Co-op report

The recovered volume is tiny relative to the area worked because the current regime is **subsidised thinning, not mechanised harvest**. That is precisely the gap the project closes.

3. Real precedents we can check

Village / Town	Forest	Harvest model	Energy result
Shimokawa, Hokkaido (下川町)	4,500+ ha town forest (FSC 7,150 ha)	**~50 ha/yr** harvest **+ replant**, 60-yr sustained cycle	56% local heat self-sufficiency; −20% town CO_2
Nishiawakura, Okayama (西粟倉村)	93–95% forest	**~3,000 ha** under 100-yr management contracts (village manages on owners' behalf)	forestry-jobs revival
Maniwa, Okayama (真庭市)	city-scale	**148,000 t/yr** (90k unused wood + 58k mill residue)	**10 MW**, 79.2 GWh/yr, ~22,000 homes, 95% uptime
Mishima, Fukushima (三島町)	~88% forest	~750 t/yr	≤50 kWe — our model's calibration anchor

Shimokawa is the closest sustained-cycle model: a whole town forest run on **~50 ha/yr by a professional, mechanised crew**. It shows what a scaled-up Mitsue Kanko can look like.

4. What the workforce can realistically deliver

Harvest volume — and therefore plant size — is set by how large and how mechanised the crew becomes. Figures below are from the project's calibrated **Forest Twin** model (sugi 600 m³/ha, 314 kg/m³, 18.5 GJ/t; net electrical efficiency 0.13, calibrated so a Mishima-scale unit reproduces ~50 kWe on ~750 t/yr). All harvested wood is routed to fuel; conversion runs over 25 years.

Cooperative capacity	Harvest rate	Area in 25 yr	Wood / yr	Biomass plant	Electricity	Heat
Current + light mechanisation	~70 ha/yr	1,762 ha (25%)	10,500 m³	~290 kWe	~2,000 MWh/yr	~6,900 MWh-th/yr
Double crew + harvester/forwarder ✅	**~140 ha/yr**	**3,526 ha (50%)**	42,000 m³	**~1,150 kWe**	**~8,000 MWh/yr**	~27,70 MWh-th/yr
Triple crew, full mechanisation	~200 ha/yr	~~5,000 ha~~ (70%)	~60,000 m³	~1.6 MWe	~11,000 MWh/yr	~38,00 MWh-th/yr
(Optimistic 90% — needs ~4× crew)	*254 ha/yr*	*6,346 ha (90%)*	*135,000 m³*	*3.7 MWe*	*~26,000 MWh/yr*	*~89,00 MWh-th/yr*

✅ **Recommended baseline: the doubled-workforce case — ~50% of the forest, ~1.1–1.2 MWe.**

5. Recommended plant

Build the plant as **2 × ~0.6 MWe units**, not one 1.2 MWe unit:

- **Redundancy / maintenance** — one unit can be serviced while the other runs.
- **FIT** — both stay under the 2 MW unused-material (未利用材) band, capturing the higher tariff (~¥40/kWh vs ~¥32/kWh for ≥2 MW).
- **Phased growth** — install unit 1 as the crew doubles, unit 2 as it approaches triple. The plant grows in step with the workforce.

Indicative capex at this scale ≈ ¥0.8 billion (¥700k/kWe installed). Electricity → grid or salt-battery thermal store; the recovered heat (~27,700 MWh-th/yr) also charges the store or supplies district heat.

Reference equipment — makers (domestic only)

Project requirement: the CHP unit must be a Japanese manufacturer — for domestic service/parts, simpler FIT + 交付金 paperwork, and to reinforce the local supply-chain / endogenous-development story with the village and 御杖村森林組合. Small woody-gasification units mostly top out at ~200–500 kWe, so ~0.6 MWe is a **cluster of modules**.

Maker	Tech / scale	Fit	Notes
中外炉工業 (Chugai Ro)	rotary-kiln gasification cogen	✅ **lead**	Since 1997, NEDO-backed; reaches our per-unit size; tolerates rough fuel (branches/bark)
神鋼環境ソリューション (Shinko / Kobe Steel grp)	~500 kWe-class up	backup	⚠️ confirm **gasification** vs steam (big plants are combustion)
静岡製機 (Shizuoka Seiki)	small ~200 kWe outdoor unit	cluster	METI-funded, aimed at agriculture/forestry — closest in spirit
ネオナイト (Neonite)	downdraft two-stage	cluster	Recovers charcoal + wood tar (biochar angle)

The **JWBA list** (`小規模木質バイオマス発電機器の一覧`) catalogues all domestic units with specs and is the single best reference. Vendor/site visits are being pursued — see `mitsue_biomass_visit_request_emails.md`.

> **Imported units are benchmark-only, not for procurement.** FORTES Energy (Latvia; 0.14–1.2 MWe gasification CHP, new Tokyo entity), Spanner Re² and Burkhardt (Germany) match the size well and are useful for spec comparison, but do not meet the Japanese-manufacturer requirement.

6. Fuel handling & drying — equipment and cost

Site the plant at the existing **Mitsue Kanko processing centre (牛峠工場, 神末797)**, which already chips and dries wood, so the CHP's heat feeds the dryers directly (no heat transport) and closes the loop **CHP heat → dry chips → CHP fuel**.

Fuel prep uses a two-stage regime: **(1) free roadside log seasoning (3–6 months, 55%→~35% moisture)**, then **(2) a short active polish-dry on CHP waste heat (~1–2 h, →10–15% for the gasifier)**. Drying load ≈ **9,000 MWh-th/yr**, about one third of the CHP's recovered heat.

Equipment and planning-grade cost (Japan, sized to ~13,000 dry-t/yr):

Stage	Equipment	Indicative cost (¥M)
Roadside seasoning	Log covers/tarps, staging area	0.5–2
Chipping	Drum/disc chipper (~10–20 t/h) *(existing 牛峠工場 may cover)*	5–30
Passive store	Ventilated covered chip barn (~500–1,000 m²)	15–40
Active polish-dry	Belt / rotary drum dryer (~2–3 t/h water, CHP-heat fed)	30–80
Screening	Sieve / screen deck	3–8
Handling	Walking-floor bin, augers, conveyors	10–25
Yard	Wheel loader / telehandler	8–20
Instrumentation	Moisture meters, controls, weighbridge	3–10
	Fuel-prep sub-total (excl. CHP gensets)	**~75–215 (mid ~120)**

CHP gensets (2×0.6 MWe) are a separate ~¥800M. **Two things cut the fuel-prep bill:** the existing 牛峠工場 kit may be sunk cost (−¥30–70M), and buying the dryer + screening

- handling **inside the gasifier vendor's package** (as Spanner Re²/Burkhardt supply in Japan) guarantees the heat/moisture match. These are scoping figures — firm numbers need a 牛峠工場 inventory audit and a vendor quote once the unit size is fixed.

> **Budget placement:** this fuel-prep capex and the CHP fleet are **Phase 4 (Operations & Scale)** items, funded by operating revenue + grants — **outside** the Phase 0–3 EVM Performance Measurement Baseline (BAC ¥220M). A small pilot-scale drying line legitimately sits inside Phase 3 (WBS 5.6 forestry ops). See `mitsue_evm_plan.md` §14.

7. Specialty broadleaf revenue — the Tenkawa model

The 25-year plan converts sugi monoculture to native broadleaf (caveat §8.1). That replant does **not have to be a cost centre**: the neighbouring Yoshino village of **Tenkawa (天川村)**, ~70 km south, has turned the same conversion into a **second revenue stream** by replanting *marketable* broadleaf instead of generic forage species. This directly answers the "why replant when sugi doesn't pay" problem and is more palatable to conservation partners (e.g. more trees) than an energy-only harvest.

Tenkawa Forest Power Council (天川村フォレストパワー協議会) — formed Dec 2016 by the **village + forest cooperative + chamber of commerce** (the same three-body structure as 御杖村 + 御杖村森林組合). On clear-cut sugi land they replant high-value natives with existing markets:

Species	Product / market	To harvest
キハダ Kihada (Phellodendron amurense)	bark for *Daranisuke* 陀羅尼助, a 1,300-yr-old herbal medicine — currently imported into the prefecture	~20 yr
ホオノキ Hoonoki (Magnolia obovata)	朴葉 leaves for sushi wrapping	—
クロモジ Kuromoji (Lindera umbellata)	aroma / essential oil	—
ウルシ Urushi	lacquer	—
maple, mizunara oak, horse chestnut, katsura	timber / craft / biodiversity	—

Why this is the same conversion pulse, monetised twice: the sugi harvest funds the crew + CHP (§4); the *replant* — instead of pure forage — is chosen for a downstream product, so the 5–20 yr gap before the new forest matures throws off **byproducts and craft income** rather than only cost. Kihada bark localises a supply Tenkawa currently buys out-of-prefecture.

Enablers Tenkawa proves work here (same terrain, same deer pressure):

- **Funding:** 農林水産業みらい基金 (Mirai Fund) grant covers nursery, forest roads and deer fencing — a template beyond the MoE 交付金 ladder.
- **Deer mitigation:** 12 m reinforced-net patch enclosures (3 m triangles failed) — directly reusable for Mitsue's replant.
- **Nursery + tending labour:** run as corporate/eco-tour volunteer days (more trees ran 育林・育苗 days there May 2024 with corporate volunteers) — an income + engagement channel that also feeds the school / eco-tourism programme.
- **Value-add:** partner Pony no Sato Farm runs bark-stripping, natural-dyeing and woodworking workshops and plans kihada-heartwood furniture — i.e. the products stay local.

Training route for Mitsue Kanko: Tenkawa also runs the **天川村森林塾 (Tenkawa Forest School)** — an 8-day felling/chainsaw course that has trained **60+ people since 2017, explicitly**

"from inside AND outside the village." It is the concrete peer-village channel for up-skilling the doubled Mitsue Kanko crew (§1). Contact: ten.forestpower@gmail.com / 0747-63-0321.

> **Action:** open a knowledge-exchange with the Tenkawa Forest Power Council and森林塾, and select **1–2 marketable broadleaf species** (kihada is the obvious lead) to fold into the replant spec so the conversion has a product, not just forage, behind it. The NGO still owns final species choice for wildlife (see reforestation principle).

8. Honest caveats

1. **Sustainability depends on the regime.** One-way conversion (clearfell → broadleaf) is a finite fuel pulse; a **rotation / mixed** regime keeps the plant fed indefinitely. The 25-year plan should be framed as a deliberate *sugi-monoculture → native-forest conversion*, funded by the energy pulse, with replanting for wildlife forage and watershed protection.
2. **Accessible area is unknown.** Steep roadless stands and fragmented private ownership (>90% private — Nishiawakura's and Mishima's #1 barrier) mean the "harvestable" fraction must be measured, not assumed. A drone + LiDAR survey (as used in Mishima) is the tool, and doubles as a trust-building gift to landowners.
3. **The crew is the constraint, not the trees.** Every figure above scales with workforce and mechanisation — the project's central investment.

Funding pathways for the Phase-4 CHP fleet (beyond the MoE 再エネ交付金 ladder): Two FY2026 national subsidies fit the biomass-power-plus-compute model directly — **経産省 GX地域共創補助金** (¥210B pool; funds *decarbonized power* + *DC/factory* as one project; Round 1 Jul–Sep 2026) and **環境省 データセンターのゼロエミッション化・地域共生加速化事業** (≤¥1B/project on DC-linked renewables + storage; FY2026 window closed 2026-07-03 — aim for FY2027). Both are logged for Baseline Rev 2 in the EVM plan §12.

9. Our reforestation values — the more trees model

The replant is not just fuel logistics or byproduct revenue (§7); it is where the project earns its ecological legitimacy. We adopt the practices of **more trees** (the forest-conservation NGO founded 2007 by Ryuichi Sakamoto with Kengo Kuma as representative director) as our explicit model — the same NGO already active in **Tenkawa** and in **Odai/Miyagawa** (§7), so its methods are proven on our terrain, deer pressure, and cooperative structure. Five values carry directly into the Mitsue replant spec:

1. **Local-provenance seedlings (地域性苗木).** Collect seed from trees native to Mitsue's own slopes, raise them, and replant them *on the same land* — never generic or out-of-region nursery stock. This protects the local gene pool and is more trees' flagship 2025 practice. It also makes the nursery itself a school / eco-tour activity (feeds the tree-survey programme).

2. **"Diverse forest-building" (多様な森づくり).** Design for biodiversity — a mix of native species *and* the animals that live in them — not another single-species plantation. No universal recipe: the species mix and method are chosen per site with 御杖村森林組合's field knowledge plus expert advice.
3. **Assisted natural regeneration + 自然配植 (natural planting-arrangement).** Where native saplings already exist, nurture them (release-cutting, ground scarification, sasa control) instead of clearing and replanting — cheaper and better suited to steep, roadless stands (§8.2). Arrange the diverse mix using the surrounding vegetation as the guide, so the result reads as a natural forest that stabilises soil and restores biodiversity. Miyagawa Forestry Cooperative (Odai) has honed this technique since 2007 with ~130 local-provenance species and teaches it across regions.
4. **Livelihoods first, city ↔ forest circulation.** Method choices must keep the local crew economically sustained. Value flows *back* to the forest through Corporate-Forest sponsorships, tours, office seedling kits (JUBAKO), wood products, and **forest-derived carbon offsets (J-Credit)** — a funding channel we should open alongside the energy revenue (§4) and Phase-4 subsidies (§8).
5. **People and time carry the cycle.** "Plant, raise, use, plant again" is carried by the hands and time of people who live here — with schools (seed→tend→plant by grade) and residents as participants, not spectators.

> **Governance:** the NGO retains final species choice for wildlife and ecosystem value (see reforestation principle); §7's marketable-broadleaf list (kihada etc.) sits *inside* these values, not against them — a diverse, native, local-provenance forest that also happens to yield craft and medicine byproducts.

Sources / 出典

- 御杖村森林組合 report (village forest ~7,051 ha; ~66 ha/yr thinning; ~120–300 m³/yr; crew avg age ~33) — `Docs extra/御杖村森林組合 (Mitsue-mura Forest Owners' Cooperative) Report.md`
- Ooba, Nakamura & Togawa (2020), *Promoting Local Revitalization to Solve Issues on Degraded Forests in Japan* — NIES Fukushima; Mishima Town CHP (≤50 kWe / 750 t/yr).
- Forest Twin model — `Docs extra/CHP biomass/forest-twin/forest_model.py` (calibrated to Mishima; all assumptions inline).
- Shimokawa Town 循環型森林経営 — https://www.town.shimokawa.hokkaido.jp/jigyo/2020/01/post-10.html
- Shimokawa Town, MAFF interview — https://www.maff.go.jp/j/shokusan/renewable/energy/interview/shimokawa.html

- Nishiawakura Village 百年の森林構想 — https://www.vill.nishiawakura.okayama.jp/wp/百年の森林構想/
- Maniwa biomass (Renewable Energy Institute, 2017) — https://www.renewable-ei.org/activities/column/20170620.html
- Tenkawa Forest Power Council (キハダ specialty-broadleaf replant, Mirai Fund) — https://www.miraikikin.org/activities/forestry/tenkawa.html
- more trees 育林・育苗 in Tenkawa (May 2024, corporate volunteers, deer-net enclosures) — https://www.more-trees.org/news/20240529/
- more trees Annual Report 2025 (local-provenance seedlings, diverse forest-building, assisted natural regeneration, 自然配植, J-Credit offsets; Odai/Miyagawa ~130 species) — `Docs extra/more_trees_annual_report_2025_EN.docx` · https://www.more-trees.org
- 天川村森林塾 (Tenkawa Forest School; 60+ trainees since 2017, open to outsiders) — https://tenforestpower.wixsite.com/tenkawajyuku
- 経産省 GX地域共創補助金 (脱炭素電源地域貢献型投資促進事業; ¥210B, DC+電源一体支援) — https://sustainablejapan.jp/2026/05/30/meti-gx-local/126036
- 環境省 データセンターのゼロエミッション化・地域共生加速化事業 (≤¥1B/件; 締切 2026-07-03) — https://sustainablejapan.jp/2026/06/07/japan-datacenter-renewable-energy/126312